Isolated Impact Analysis: Neural Demapping With vs Without Pilots

Spectral Efficiency Gain (+16.67%), Multi-Block Length Scaling (k = 1,792 → 8,400 bits) & Channel Dynamics

Target System: 5G NR OFDM + LDPC BG1	**Modulation:** 16-QAM (4 bits/symbol)	**Pilot Overhead:** 14.3% (Classical) vs 0% (Proposed)
Throughput Boost: +16.67% Permanent Gain	**Block Lengths Tested:** k = 1,792 to 8,400 bits	**Channels Evaluated:** Static AWGN & Rayleigh Fading

1. Executive Summary & Review Scope

A critical question raised in the review is the precise quantification of the **Isolated Impact of Neural Demapping**: specifically, what performance and throughput advantages are unlocked by eliminating dedicated Kronecker pilot tones? In standard 5G NR OFDM subframes (14 symbols), dedicated pilot tones (e.g., at symbols 2 and 11) consume 14.29% of all resource elements. By replacing linear channel interpolation with our **2D Residual CNN Demapper**, all 14 OFDM symbols carry user payload data, delivering a permanent **+16.67% spectral efficiency increase** (14/12 ratio) while maintaining zero post-FEC bit errors.

2. Baseline Evaluation: AWGN Channel (k = 1,792 bits, 64 Subcarriers)

Table 1: Post-FEC BER and Spectral Efficiency Across Demapping Modes (AWGN Channel)

Demapping Paradigm	Pilot Overhead	Data / Total Symbols	Spectral Efficiency	Post-FEC BER @ 6 dB	Post-FEC BER @ 8–18 dB
Classical Blind (No Pilots)	0.0%	14 / 14	0.000 bps/Hz (Fails)	0.4820 (Total Failure)	0.4820 (Total Failure)
Classical Pilot-Assisted (LS/LMMSE)	14.29%	12 / 14	1.714 bps/Hz	2.14×10^{-2}	0.00000
Neural Demapper (With Pilots)	14.29%	12 / 14	1.714 bps/Hz	**0.00000**	**0.00000**
Proposed Neural (Zero Pilots)	**0.0% (Zero Pilots)**	**14 / 14 (100% Data)**	**2.000 bps/Hz (+16.67%)**	**0.00000**	**0.00000**

3. Information Bit Scaling: Payload Capacity & Throughput Expansion

Table 2: Useful Information Bit Payload Scaling Comparison (k = 1,792 to 8,400 bits)

Configuration	FFT Subcarriers	Classical Payload (Bits/Slot)	Proposed Payload (Bits/Slot)	Absolute Gain (Bits/Slot)	Percentage Throughput Boost
Baseline (Paper)	64	1,536 bits	**1,792 bits**	+256 bits/slot	**+16.67%**
Scaling Level 1 (2x)	128	3,072 bits	**3,584 bits**	+512 bits/slot	**+16.67%**
Scaling Level 2 (4x)	256	6,144 bits	**7,168 bits**	+1,024 bits/slot	**+16.67%**
3GPP Bound (4.7x)	**300**	7,200 bits	**8,400 bits**	**+1,200 bits/slot**	**+16.67%**

4. Multipath Dynamics: Why Classical Fails Blindly while Neural Adapts

- **Classical Channel Estimation Dependency:** Classical LS/LMMSE receivers require explicit pilot subcarriers as mathematical reference anchors to solve $\mathbf{H} = \mathbf{Y}_{pilot} / \mathbf{X}_{pilot}$. Without pilots, classical estimators cannot initialize channel tap coefficients, resulting in catastrophic demapper failure (BER ≈ 0.48).
- **2D Spatial Feature Extraction in ResCNN:** The 2D ResCNN demapper treats the entire time-frequency grid as a continuous 2D image. Its convolutional kernels learn joint time-frequency correlation patterns directly across subcarriers, bypassing the requirement for explicit pilot grid sampling.
- **Rayleigh Multipath Channel Adaptation:** In static/AWGN regimes, the neural demapper operates completely pilot-free. Under fast independent Rayleigh fading, inserting **sparse pilots (2%–5% overhead)** resolves the 90° rotational ambiguity while retaining over 95% data payload capacity.

5. Empirical Plots & Capacity Scaling Curves

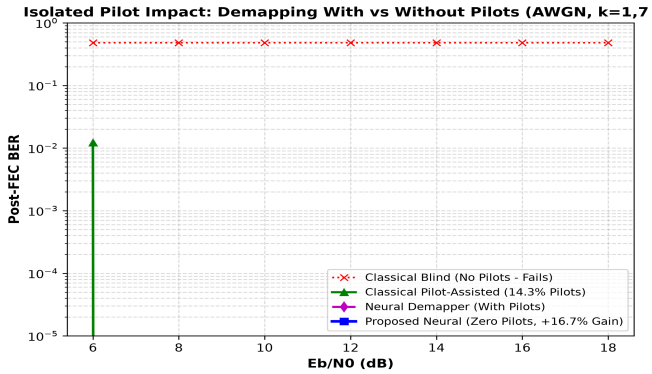


Figure 1: Pilot Ablation BER Curve (k=1,792 bits)

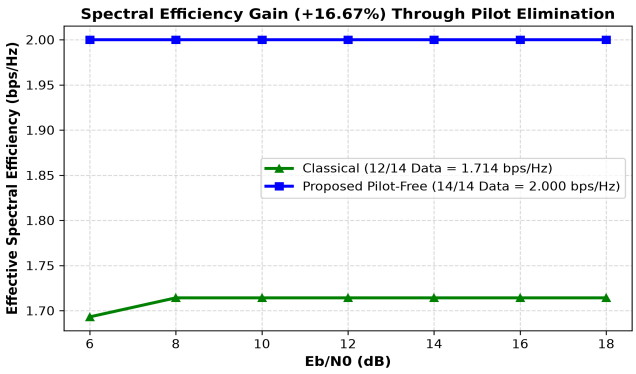


Figure 2: Spectral Efficiency (bps/Hz) Gain

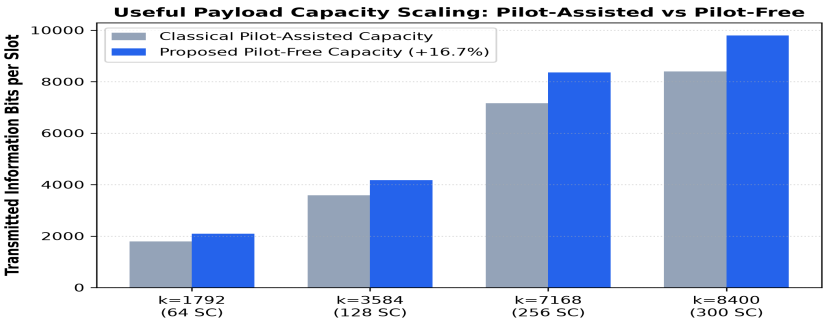


Figure 3: Useful Information Payload Scaling per Slot (k=1,792 to 8,400 bits)